

Gabriella Sekar Shada

Computer Science Educator and Researcher | Speciality in Robotics, HCI, Haptic, Mixed Reality

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SUMMARY

Computer Science educator and researcher with a Master's (Distinction) from the University of Melbourne and 5+ years of industry experience. Passionate about delivering meaningful teaching and learning experiences both in the classroom and through research, particularly in the intersection of technology and healthcare. Experienced in mentoring, curriculum design, and student-centred learning, with a commitment to nurturing students' growth in knowledge, character, and purpose.

EDUCATION

Master of Computer Science (with Distinction)

Dec 2025

The University of Melbourne

Melbourne, Australia

WAM: 78.438 / 100. Specialisation in Human Computer Interaction

Thesis: Exploring Mixed Reality, Haptic Feedback, and Social Robots to Advance Patient-Centric Physiotherapy Training (25,871 words) [\[GitHub\]](#) [\[PDF\]](#)

- Thesis ranked in the Top 10%, Department of Computer Science, University of Melbourne (Dec 2025)
- Curator's Choice Award, Melbourne Design Week. Awarded for research-based interaction design project in Designing Novel Interactions (INFO90003) (May 2024)
- Top 5 Project in Research, Innovation and Commercialisation Awards, [Endeavour Exhibition](#), University of Melbourne (2025); featured in a student research showcase by the University of Melbourne. <https://vt.tiktok.com/ZSuAX6GQf/>
- Fully funded by Indonesia Endowment Fund for Education (LPDP) Scholarship. A competitive Master's degree scholarship from Indonesian government for high achieving graduate study candidate abroad.

Sydney XR Research Methods Summer School

Jan 2025

The University of Sydney

Sydney, Australia

Bachelor of Applied Computing, Mobile Communication and Technology (GPA 3.45)

Sept 2018

Sampoerna University

Jakarta, Indonesia

Research Project: An Android Based Mobile Library Application for Sampoerna University (12,585 words)

Associate of Arts (Double degree with Sampoerna University), *Cum Laude* (GPA 3.56)

Dec 2016

Lone Star College

Jakarta, Indonesia

RESEARCH OUTPUTS

Shada, G. S., Andaru, L., Khor, K., Wald, B., & Huang, M. (2024). NISCALA – Converging human emotions with interactive technologies. In *Proceedings of the International Conference on Computer Simulation (ICoCSIM)*, 250–253. <https://doi.org/10.1109/ICoCSIM65098.2024.00051>

Contribution: Led the interaction and visual design of the NISCALA system, including interface flow, aesthetic direction, and laser-printing fabrication design. Contributed to paper writing and presentation (approx. 40% overall contribution.)

Shada, G. S., & Anugerah Ayu, M. (2018). Designing Android user interface for university mobile library. In *Proceedings of the International Conference on Computer Engineering, Design and Technology (ICCED)*, 224–229. <https://doi.org/10.1109/ICCED.2018.00051>

Contribution: Led the project end-to-end as primary researcher, conducting fieldwork interviews with university librarians to establish requirements, designing and iterating low-fidelity sketches and prototypes, and evaluating the interface with users through usability testing. Authored the majority of the paper.

RESEARCH EXPERIENCE

Social Robot and Visualisation for Medical Training

Master's Thesis in Interactive Technology Lab (IXT)

Feb 2025 – Dec 2025

- Collaborated **interdisciplinary** with the Department of Physiotherapy to **design and iterate** on a reliable MR training system, conducting expert **interviews** to inform system requirements and refinement.
- Developed a **Unity**-based mixed-reality research platform integrating **Furhat Robot APIs**, ESP32 **micro-controller Bluetooth sensors**, and immersive visualisation to support **multimodal interaction** experiments.
- Integrated **Manus motion-capture** gloves and configured **OptiTrack (Motive)** for real-time motion tracking.
- Built experimental prototypes in Unity, interfacing with hardware and Furhat robotics systems from **Wizard-of-Oz** to **API integration**.
- Designed and conducted user studies combining **NASA-TLX** cognitive load assessment, **think-aloud** protocols, **questionnaire**, and **participatory design** to evaluate how physiotherapy students made sense of a multimodal MR system.

Sepsis Prediction from ICU Clinical Data

Machine Learning Applications for Health (COMP90089), The University of Melbourne

Aug 2024 – Nov 2024

- Built an end-to-end clinical ML pipeline predicting sepsis onset in ICU patients using time-series EHR data from the **MIMIC-III** database (40,000+ ICU stays).
- Engineered features from vital signs, SOFA scores, and physiological variables; handled missing data and class imbalance across train, validation, and test splits.
- Trained and evaluated deep learning models (**LSTM**, **GRU**) and ensemble classifiers (AdaBoost, Random Forest) using **PyTorch**; best model (LSTM) achieved **ROC-AUC of 0.917**.
- *Research-relevant:* Gained hands-on experience in deep learning for sequential clinical data, real-world electronics health records feature engineering, and systematic model evaluation and analysis.

PROFESSIONAL EXPERIENCE

Senior Front End Developer

[Alodokter](#)

Jun 2019 – Jan 2024

Jakarta, Indonesia

- Architected and rebuilt insurance product pages, doctor profiles, and article platforms with full backend integration in **Ruby on Rails**, serving **25M+** active users.
- Refactored legacy **Polymer** frontend codebase to boost PageSpeed scores from 20 to 70, achieving over **100% performance** improvement, alongside enhancements to image rendering and SEO.
- Delivered the **Aloproteksi frontend end-to-end**, a corporate insurance dashboard built with Polymer and integrated with backend services via Ruby on Rails APIs, to scale online insurance sales operations.
- *Research-relevant:* Developed practical experience in systematic tool evaluation, iterative design under technical constraints, and assessing how design decisions affect user-facing outcomes at scale — directly informing my later research approach in human-centred system design.

Programming Mentor

[Eduwork](#)

Nov 2021 – Feb 2022

Yogyakarta, Indonesia

- Delivered weekly online mentoring sessions in **MERN stack**, provide code reviews and technical feedback.
- Guided students in building portfolios to prepare their industry roles through hands-on project supervision.
- Facilitated problem framing and brainstorming sessions to help students identify meaningful project ideas, guiding them toward portfolio work that was not only technically functional but socially relevant and impactful.
- *Research-relevant:* Gained early experience in participatory problem scoping and translating real-world needs into design briefs as the foundation of their fullstack role. This process closely aligned with co-design and requirements-gathering methodologies used in HCI research.

Python Trainer

Sampoerna University

Aug 2021

Jakarta, Indonesia

- Designed a targeted **Python** syllabus and hands-on lesson plans to prepare the first year university students for the Microsoft Technology Associate (MTA) certification.
- Managed classroom resources and ensured consistent student access to learning materials, adapting delivery to fully online formats during the COVID-19 pandemic.

Software Engineer

Edge Property Singapore

Nov 2018 – May 2019

Jakarta, Indonesia

- Collaborated with a cross functional team to redesign property listing pages in **React**, improving click-through and user engagement metrics across the platform.
- Built and owned two agent-facing features including pro-tools and pro-points from frontend to backend, helping agents manage listings more efficiently and close more sales.

HONOURS & AWARDS

- **Honourable Mention** in CISSA Design Blitz May 2025
- **First Place in UX Design**, CISSA Design Blitz, The University of Melbourne April 2024
- **Top 15 from 50+ teams**, FEIT Hackathon Festival, The University of Melbourne 2024 & 2025

COMMUNITY & ACADEMIC ENGAGEMENT

- **Moderator**, Plenary Session "Emerging Technology, Engineering & Digital Innovation", 10th Indonesian Students' International Conference July 2025
- **Girl Power Mentor** in STEM for high school student, The University of Melbourne 2025
- **Student Panellist & Team Mentor**, FEIT Orientation Week, The University of Melbourne July 2025
- **Vice President**, LPDP University of Melbourne Student Association 2024–2025
- **Guest Speaker**, Sampoerna University Alumni Sharing 2021

PERSONAL DETAILS

- Languages: Indonesian (Native), English (Professional/Fluent) — IELTS 7.0
- Nationality: Indonesian